

Risk Revisited*

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ABSTRACT

This study explores how stock price path shapes investors' risk perceptions. In four experiments, we present participants with real and fabricated price charts and elicit risk perceptions. We document that a parsimonious set of three features – recency, clustering, and sign – can explain large proportions of risk perception variations. We find that the effects are partially mediated by perceived volatility, which is not fully explained by Bayesian inference. Additionally, a significant portion of the effects occurs independently of perceived volatility. Using observational data, we show that these three features explain cross-sectional variations in stock returns and predict mutual fund negative flows.

JEL Classification: C90, G40, D80

Keywords: Risk perception, stock market, experiment, recency, cluster, sign

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1 Introduction

Understanding the risks associated with financial assets is central to financial decision-making. Since [Markowitz \(1952\)](#), many theoretical studies have conceptualized asset risk as return volatility, often implemented in empirical studies as the standard deviation of realized historical returns.¹ While this definition is theoretically appealing and empirically convenient, the perception of risk as an input to financial decisions is inherently subjective. In this paper, we experimentally study how features of an asset’s historical price path affect investors’ perceptions of its riskiness.² We then use observational data to illustrate how these effects help us understand investment decisions and asset prices in real financial markets.

We propose that risk perceptions of an asset, holding constant the second and higher-order moments of the realized return distribution, are influenced by three key features of its historical price path: recency, clustering, and sign.³ Recency captures the extent to which recent returns are more volatile than earlier ones. Clustering quantifies the degree to which consecutive returns of the same sign create less but larger fluctuations. Sign represents the average return over the sample period, reflecting the overall direction of the price path.

While the three features we study may share potential underlying mechanisms (e.g., limited attention), their empirical correlations are weak, suggesting they capture distinct dimensions. The three features find their roots in psychology and behavioral economics literatures. For example, [Malmendier and Nagel \(2016\)](#) and [Nagel and Xu \(2022\)](#) show that investor beliefs and decisions disproportionately weight recent experiences relative to distant ones, highlighting the role of recency in shaping risk perceptions.⁴ Moreover, [Gulen and Lim \(2024\)](#) demonstrate that larger fluctuations are more visually salient, implying that

¹Risk is also often defined as an asset’s return covariance with systematic factors, which is the non-diversifiable part of the return volatility. Nevertheless, [Chinco et al. \(2022\)](#) show that investment decisions of individual investors primarily depend on the total volatility.

²We emphasize historical price information because it is the most widely accessible type of information for investors.

³We initially named clustering “representativeness” in the pre-registration, but changed to the current name to avoid confusion with the representativeness in the belief updating literature.

⁴Relatedly, [Greenwood and Shleifer \(2014\)](#) and [Lochstoer and Muir \(2022\)](#) find that investors’ expectations of returns and volatility overreact to recent news.

sequences of same-sign returns forming pronounced swings are more likely to be perceived as risky. Finally, prospect theory (Kahneman and Tversky, 1979) emphasizes that losses loom larger than gains, suggesting that fluctuations within downward trends should amplify perceived risk.

To study the effects of these features on risk perceptions, we conduct a series of experiments where participants evaluate the riskiness of assets and make investment decisions after observing their historical price paths. In the main experiment, we present participants with 12-month price paths of both artificially-generated assets and real U.S. stocks. The artificial assets' price paths are carefully designed: the distributions of monthly returns are identical across all artificial assets up to a constant shift while the sequences of returns are reshuffled. This experimental design enables us to systematically manipulate the recency, clustering, and sign of the price paths while holding constant the second and higher-order moments of the return distributions. Moreover, while participants do not know which assets are artificial and which ones are real stocks, they are informed that their bonus payments depend on their answers for a randomly chosen real stock. This incentive structure motivates participants to evaluate all assets as though they were real, thereby enhancing the external validity of our findings. For external validity, in a separate experiment, we also present subjects with real stock price paths without holding the distribution of returns fixed.

We find that higher recency, increased clustering, and lower sign values consistently heighten participants' risk perceptions and reduce their willingness to invest after controlling for subjective expected returns. In additional experiments, we show that these effects are robust to the frequency of price data and the length of the sample period. They also persist regardless of whether these features are manipulated one at a time or simultaneously.

To investigate the mechanisms of these effects, in the main experiments, we also ask participants to predict the assets' future prices and elicit their confidence in these predictions using an incentive-compatible method. The confidence elicitation gives us a measure of participants' perceived volatility, allowing us to examine whether the impact of the three

price path features on risk perceptions operates through their influence on perceived volatility. Our findings indicate that the effects of recency, clustering, and sign on risk perceptions are partially mediated by their impact on perceived volatility. However, a significant portion of these effects occurs independently of perceived volatility, directly influencing risk perceptions.

To further investigate whether the effects of price path features on perceived volatility is consistent with Bayesian inference, we conduct an additional experiment in which participants are presented with artificial assets and have full knowledge of their return-generating processes. Under these processes, Bayesian participants’ perceived volatility should not depend on the three features we vary. Contrary to this Bayesian benchmark, we find that the effects of recency, clustering, and sign persist—both in terms of perceived volatility and risk perceptions. This result further confirms that the effects of these features on risk perceptions cannot be fully explained within a Bayesian framework.

While our analysis focuses on the effects of three specific features, we recognize that risk perceptions may be influenced by additional characteristics of price paths (see the literature review below). Nonetheless, our three-feature model offers a notable advantage: it covers conceptually and empirically distinct aspects of price path patterns with minimal complexity. The parsimony of our model comes with little cost in explanatory power. Using our experimental data, we show that our three-feature model accounts for 80% of the explainable variation (as defined in [Fudenberg et al. \(2022\)](#)) in participants’ risk perceptions. This explanatory power is more than double that of a benchmark model based solely on objective volatility and is close to a more complex nine-feature model from the literature.

The three features in our model are designed to be readily portable to real data applications — they are all homogeneous of degree 0 with respect to price level and sample horizon. To demonstrate the usefulness of our three-feature model for explaining asset prices and investment decisions in real financial markets, we analyze observational data on U.S. stock returns and mutual fund flows. For stocks, we construct long-short, monthly rebalanced portfolios based on each of the three features. We find that a high-minus-low (low-minus-

high) portfolio for the recency or clustering (sign) feature—corresponding to higher perceived risk—exhibits lower returns in the subsequent month. These findings are consistent with the effects of the three features on risk perceptions found in our experiments.⁵ For mutual funds, we adopt fund flow measure as defined by [Coval and Stafford \(2007\)](#) and observe that higher recency or clustering, as well as lower sign, increases the likelihood of negative fund flows. These results indicate that the effects of risk perceptions on investment are not restricted to lab settings and small stakes; instead, they appear to be affecting people’s real-world financial investments as well.

Our paper builds on the literature that studies people’s risk perceptions of stock price charts. [Grosshans and Zeisberger \(2018\)](#) show that the patterns of price charts affect investors’ risk preferences, price forecasts, and subjective well-being. [Duxbury and Summers \(2018\)](#) find that price-based factors play a large role in determining investors’ risk perceptions. [Bose et al. \(2022\)](#) apply a machine learning algorithm and find that the visual salience of different parts of a price chart affects perceived risk. [Borsboom and Zeisberger \(2020\)](#) use a similar experimental setting and find that price chart features such as maximum price, minimum price, and crashes are important drivers of perceived risk even after controlling for the distribution of historical returns. Moreover, [Nolte and Schneider \(2018\)](#) show that the effects of price chart features remain even if participants are presented with the true return distribution. Lastly, [Holzmeister et al. \(2020\)](#) compare risk perceptions of lay people and financial professionals and find them to be remarkably similar.

Our experiment builds on the design used in this literature. Following prior work, we elicit risk perceptions using self-reported risk scores. In particular, our use of price charts with reshuffled returns is inspired by [Nolte and Schneider \(2018\)](#), which presents participants with four price charts that share identical return distributions. Our three-feature model is

⁵The evidence we find with sign feature suggest that higher past return leads to higher future return in the short-run through the channel of lower perceived risk. While we do not claim causality of our risk perception channel, this explanation can be an alternative mechanism for the short-term momentum ([Jegadeesh and Titman, 1993](#)).

also informed by this line of research and our experimental findings align with earlier results.⁶ We contribute to this literature in two key ways. First, we advance the understanding of risk perception by examining its relationship with perceived volatility and expected returns, and by clarifying the psychological mechanisms through which specific chart features influence risk judgments. Second, we address issues on the broader applicability of risk perception features in financial markets. Specifically, we focus on features that are invariant to price levels and sample horizon, demonstrate that these features’ effects on risk perceptions apply to both artificial and real assets, highlight the three-feature model’s strong explanatory power despite its simplicity, and illustrate its usefulness in explaining investment decisions and asset prices in real-world financial markets.

The rest of this paper is organized as follows. Section 2 provides the design and results of our first experiment where the three features are explored independently. Section 3 and 4 cover two additional experiments built upon the first one that allow us to mix the three features and explore the mechanism. Section 5 presents the applications of our experimental findings, which are based on observational data from U.S. individual stocks and mutual funds. Finally, Section 6 concludes.

2 Experiment 1

2.1 Design

In this experiment, each participant sees eight stock price charts in random orders. Each price chart displays the path of end-of-the-month stock prices over a 12-month period, with the final price normalized to 100 (see Figure 7). We choose to present stock prices on each chart simultaneously throughout our experiments to highlight the effects of our three features under the same treatment of simultaneous information presentation.⁷ One of these charts

⁶For example, [Borsboom and Zeisberger \(2020\)](#) finds that recent price changes and volatility, sizes of crashes, and total returns affect investors’ risk perceptions.

⁷[Bohren et al. \(2024\)](#) find that learning simultaneously and learning sequentially have different implications for unlikely but salient events.

represents an actual U.S. company, while the others are fabricated. The real price chart is selected using a quasi-random method. First, we extract the end-of-the-month stock prices of all U.S. stocks from 2019 to 2020 whose return mean and standard deviation are within 20% distance to those of our fabricated benchmark returns. This approach ensures the real price chart does not appear significantly different from the fabricated ones. Then, we randomly select 100 of them to form a database. The real price chart a participant sees is randomly selected from this database.

2.1.1 Price Charts

The fabricated price charts, summarized in Table 1, are all based on the same 12 monthly percentage returns: -10, -5, -2.5, -2.5, -1.25, -1.25, 1.25, 1.25, 2.5, 2.5, 5, and 10.⁸ The returns in Benchmark, Early, Recent, and Clustered are reshuffles of these 12 numbers. In Benchmark, positive and negative returns alternate, and the magnitude of returns is symmetrically distributed around the center. Early (Recent) keeps the alternating feature between positive and negative returns but concentrate larger returns in earlier (more recent) months. In Clustered, positive returns are primarily clustered in the earlier months, while negative returns predominantly occur in the later months. In Positive, we add 1.125 to the 12 monthly returns while keeping the signs of returns alternating and the distribution symmetric around the middle. Negative is similar except that the constant 1.125 is subtracted. Volatile is Benchmark with all returns amplified by a factor of 2.

By varying the three features independently, these chart designs allow us to study their individual effects on risk perception. Comparison between Benchmark, Early and Recent can reveal the effect of the timing of return volatility. Clustered, compared with the Benchmark, allows us to identify how the clustering of same-sign returns affects perceived risk. Positive and Negative are designed to study the effect of average returns. Lastly, the comparison between Volatile and Benchmark measures the effect of return volatility, which serves as a

⁸This design is based on [Nolte and Schneider \(2018\)](#). In their experiment, participants are presented with four price charts with reshuffled daily returns.

baseline for the effect sizes of other features.

We focus on the following three features of price charts: recency is defined as

$$recency = \frac{s_3 - s_1}{\bar{s}} \quad (2.1)$$

where s_i is the i th-fold (of the three-fold-equally-split) subsample's standard deviation of returns and $\bar{s}$ is the average of s_1 , s_2 , and s_3 . Clustering is defined by

$$\begin{aligned} clustering = & \frac{\sum_{(t,t+1) \in \{(t,t+1): \forall t \leq T-1 \text{ s.t. } r_t \cdot r_{t+1} > 0\}} (|r_t| + |r_{t+1}|)}{\sum_{(t,t+1) \in \{(t,t+1): \forall t \leq T-1\}} (|r_t| + |r_{t+1}|)} \\ & \times \frac{std(\{r_t : \forall t \leq T\}) - std(\{|r_t| : \forall t \leq T\})}{std(\{r_t : \forall t \leq T\})} \end{aligned} \quad (2.2)$$

where the $std(\cdot)$ is the standard deviation operator for a set of returns. Note that the second multiplier in (2.2) regulates the measure such that it will only generate a high value if meaningful fluctuations exist.⁹ Intuitively, the clustering measure given by (2.2) assigns a high value to a price chart only if it satisfies both the same sign returns' clustering and the non-monotonicity of the price path. Sign is simply the average return, i.e.,

$$sign = \frac{1}{T} \sum_{t \leq T} r_t. \quad (2.3)$$

Note that the definitions of all three features are homogeneous of degree 0 with respect to price level and sample horizon. This makes the definitions portable to other settings without having to make ad-hoc adjustments.

The return series that we use to generate treatment price charts for Experiment 1 as well as their corresponding three feature scores are provided in Table 1.

For each price chart they view, participants are asked to answer four questions. The primary question (Risk Score) asks them to rate the stock's riskiness on a slider ranging from

⁹For instance, for a sequence of all-positive or all-negative returns, our cluster score will be zero because of this multiplier, which is consistent with the intuitive nature of risk.

Table 1: Fabricated price charts in Experiment 1

This table reports the return series for our seven treatment price charts, in which Early and Recent correspond to recency feature, Clustered correspond to cluster feature, Positive and Negative correspond to sign feature, and Benchmark and Volatile correspond to the effect of volatility as our baseline. The table also presents volatility and scores for recency, cluster and sign per (2.1), (2.2), and (2.3), respectively.

	Returns	Volatility	Recency	Cluster	Sign
Benchmark	-1.25, 2.5, -2.5, 1.25, -5, 10, -10, 5, -1.25, 2.5, -2.5, 1.25	5.056	0	0	0
Early	10, -10, 5, -5, 2.5, -2.5, 1.25, -1.25, 2.5, -1.25, 1.25, -2.5	5.056	-1.5	0	0
Recent	2.5, -2.5, 1.25, -1.25, 2.5, -1.25, 1.25, -2.5, 5, -5, 10, -10	5.056	1.5	0	0
Clustered	2.5, -1.25, 2.5, 1.25, 10, 5, -10, -5, -1.25, -2.5, 1.25, -2.5	5.056	0	0.238	0
Positive	-1.375, 2.375, -0.125, 3.625, -3.875, 6.125, -8.875, 11.125, -1.375, 2.375, -0.125, 3.625	5.056	0	0	1.125
Negative	-3.625, 0.125, -2.375, 1.375, -11.125, 3.875, -6.125, 8.875, -3.625, 0.125, -2.375, 1.375	5.056	0	0	-1.125
Volatile	-2.5, 5, -5, 2.5, -10, 20, -20, 10, -2.5, 5, -5, 2.5	10.113	0	0	0

0 to 10. Following prior literature, we do not provide an explicit definition of "risk," allowing participants to apply their own interpretations. This approach captures individualized notions of risk perception, but also introduces ambiguity regarding the specific components participants consider. To address this limitation, the remaining questions are objectively defined and serve as potential validation measures for the subjective risk scores. The second question (Investment) asks participants to allocate a \$3 endowment between the stock and a safe, zero-interest account at the end of the 12-month period. Any shares of stock purchased will be sold three months later. The third question (Forecast) asks participants to forecast the stock price three months after the end of the 12-month period. The fourth question (Confidence) asks participants how likely it is that the forecast is less than 5 point off the actual price.

2.1.2 Incentivization

Except for the question about the risk score of a stock, all other questions in the experiment are incentivized. Each participant is eligible for a bonus determined by their response to a randomly selected question concerning the real price chart.¹⁰¹¹

If the investment question determines the bonus, it will be equal to the value of the \$3 endowment at the end of the third month. If the forecast question determines the bonus, it will be \$3 if the forecast is less than 5 points off the actual stock price, and \$0 otherwise. If the confidence question determines the bonus, its calculation will utilize a Becker-DeGroot-Marschak (BDM) mechanism. Specifically, the computer will randomly generate a number between 0 and 100%. If the reported confidence is smaller than this number, the probability of receiving a \$3 bonus is equal to this number. If the confidence is greater than this number, the participant receives a \$3 bonus if the forecast is less than 5 point off the actual stock price. We advise participants that providing truthful and accurate responses to each question is in their best interest. Following the best practice recommended by [Danz et al. \(2022\)](#), the details of the incentive mechanism are provided, but participants are not required to read them.¹²

A crucial feature of our incentive scheme is that participants are incentivized to treat every price chart as if it depicts a real stock because only the real price chart matters for bonus payment. This design enhances the external validity of our experimental findings.

¹⁰Paying participants based on a random question encourages participants to treat each question in isolation and eliminates interdependence between answers that arises because of income effects or hedging motives ([Azrieli et al., 2018](#)).

¹¹Paying participants based on their answers about a real price chart has two benefits. First, it frees participants from having to second-guess how the experimenters determine the future prices of the fabricated price charts. Second, it encourages participants to treat all price charts as if they are real, making our experimental findings more likely to be externally valid.

¹²[Danz et al. \(2022\)](#) shows that this practice reduces non-optimal gaming of the incentive mechanism and encourages truthful reporting.

2.1.3 Participants

Recent studies show similar findings on risk perceptions using samples of financial professionals, MBA students, and laypeople (Holzmeister et al., 2020; Chinco et al., 2022). In our study, we recruited 100 participants from Prolific who meet the following eligibility criteria: (1) from the U.S.; (2) has completed at least 20 previous studies; (3) approval rate is at least 99%; and (4) self-reported to have investment experience.¹³ We offer a \$4 participation fee and the median duration of the experiment is 14 minutes.¹⁴

2.2 Result

Figure 1 presents the main results of the first experiment, with the upper and lower charts corresponding to our two measures of riskiness, i.e., risk score and confidence, respectively.

For the seven charts of interest,¹⁵ the figure shows the mean and 95% confidence interval for all charts. The p -value significance levels from the Wilcoxon signed-rank tests are reported for six charts (Volatile, Early, Recent, Negative, Positive, Clustered) against the Benchmark.¹⁶

The results suggest that high volatility, high recency, negative sign, and clustered charts are perceived as riskier than the benchmark chart. The high volatility chart, which doubles the volatility of the benchmark, is perceived as riskier, affirming that traditional volatility measure of risk remains valid. However, the three factors we propose in this paper are also significant. Comparing to the magnitude of change in risk score and confidence, we find high recency chart has about 21% increase in risk score and 56% decrease in confidence of those by doubling the volatility; negative sign and clustered charts have similar or even greater

¹³Prolific is a popular platform for recruiting diverse and pre-screened U.S. workers for experimental and survey studies in economics and finance (Palan and Schitter, 2018). It offers higher pay than similar platforms and has a more representative pool of participants (Bergman et al., 2020).

¹⁴Details of Experiment 1 were pre-registered and are available at https://aspredicted.org/qq7_h2g.

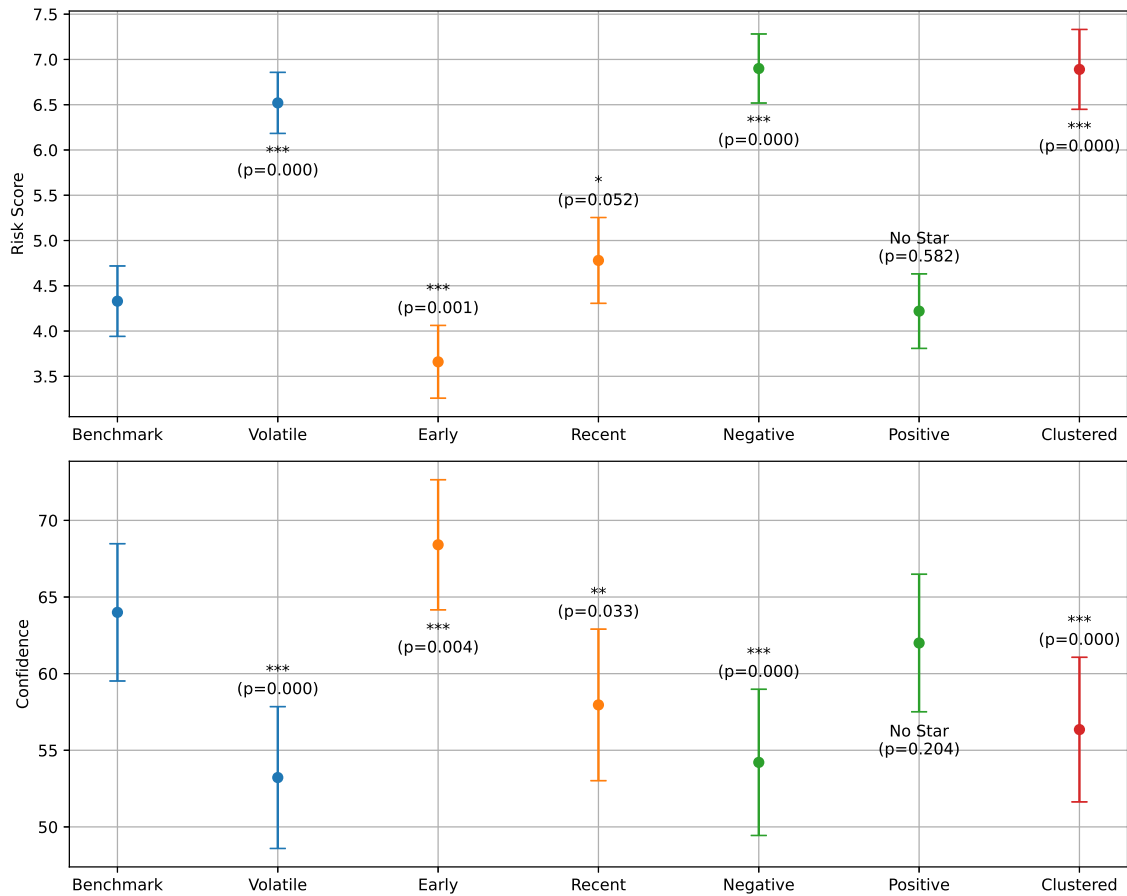
¹⁵Experiment 2 explores a larger set of real and reshuffled charts; therefore, the real chart from Experiment 1 is omitted here.

¹⁶The Wilcoxon signed-rank test is a non-parametric counterpart of the t -test, which does not assume a normal distribution.

increase (decrease) in risk score (confidence) than those by doubling the volatility. These findings, consistent across both risk perception measures, indicate that the three investigated features exert statistically and economically significant effects on risk perception.

Figure 1: Main Results of Experiment 1

The figure compares mean and 95% confidence interval for two subjective risk measures, i.e., risk score and forecast confidence, across different price chart treatments in our Experiment 1. Wilcoxon signed-rank test p -values are presented in parentheses. ***, **, and * indicate statistical significance at 1%, 5% and 10% levels, respectively.

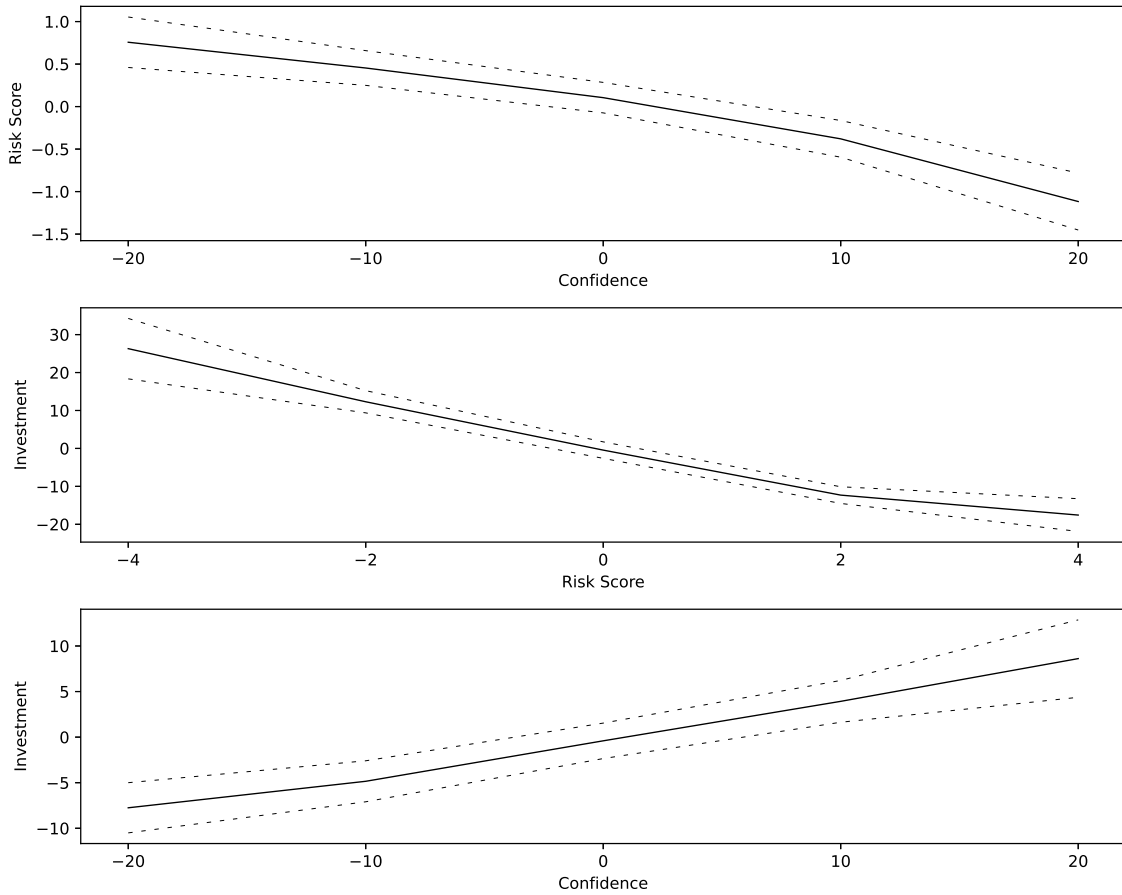


Since risk scores lack a universally accepted definition, it is important to validate our measure as a meaningful proxy for participants' perceived risk. We conduct two such validation exercises. First, although risk perception may encompass factors beyond volatility, any credible measure should at least exhibit a positive correlation with perceived volatility, which in our experiment is measured by participants' confidence in their own price predictions.

Second, to be behaviorally meaningful, a measure of risk perception should be predictive of investment decisions.

Figure 2: Perceived Risk Measures and Real Consequence

The figure shows the relations between risk score and confidence, investment and risk score, and investment and confidence. All variables are pooled across subjects and demeaned for each subject to control for the subject fixed effects. The solid lines are average values and the dashed lines represent the 95% confidence interval.



To test these correlations, we plot Figure 2, in which the upper chart depicts the relation between risk score and confidence and the next two charts show the relation between investment and each one of the two risk perception measures. All subjects are pooled in these charts, so we subtract each subject's mean answer to all 8 charts from his answer to each individual chart to control for subject fixed effects. As expected, the two risk perception measures are negatively correlated in a monotonic way. Both measures imply that higher

perceived risk monotonically leads to lower investment.

3 Experiment 2

3.1 Design

In Experiment 1, each fabricated price chart deviates from the benchmark chart in only one of the three features. To test if the effects of the three features on risk perceptions and investment decisions persist when all features are manipulated simultaneously, we run a second experiment with 500 Prolific participants. Specifically, in Experiment 2, each participant answers the same four questions as in Experiment 1 for 10 fabricated price charts and 10 real ones.¹⁷ The real charts are randomly selected from the same database of 100 real prices charts as in Experiment 1. The key departure from Experiment 1 is that the returns in the fabricated price charts of Experiment 2 are random reshuffles of the monthly returns of three charts in Experiment 1: Benchmark, Positive, and Negative. As a result, any two fabricated charts could differ significantly in more than one feature. This rich mixture of feature variations, along with a larger sample size, also allows us to investigate the mechanisms behind the features’ effects on risk perceptions.

As before, participants do not know which chart is fabricated and which is real. The bonus a participant receives is based on their response to a randomly selected question about a real price chart. Experiment 2 pays a \$5 participation fee and has a median duration of 22 minutes. The eligibility criteria for participation are the same as in Experiment 1.¹⁸

3.2 Result

Main effects. With three features mixed in all the 200 graphs ($N_{\text{graph}}^{\text{step } 2} = 200$) and more participants ($N_{\text{participant}}^{\text{step } 2} = 500$), we conduct regression analysis to understand the effects and

¹⁷The number of responses to each price chart is shown in Figure 8 in Appendix A1.

¹⁸Details of Experiment 2 were pre-registered and are available at https://aspredicted.org/Z7L_RRY.

to properly control for subject fixed effects and other control variables.¹⁹

Table 2: Risk Perception Determinants

This table reports the estimates from the ordinary least squares (OLS) regression given by (3.1) using the data from Experiment 2. For each price chart j , volatility is the standard deviation of returns; the recency, cluster and sign scores are computed by (2.1), (2.2), and (2.3), respectively. Risk score, forecast, and confidence are specific to and thus indexed by subject i , which are collect from our questions under price chart j . Standard errors are clustered on subject. Robust t -statistics are presented in parentheses. ***, **, and * indicate statistical significance at 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	Three Features		Decompose Risk Score	
	risk score	confidence	risk score	risk score
recency	0.249*** (6.95)	-1.130*** (-4.92)	0.218*** (6.13)	0.207*** (6.10)
cluster	3.268*** (7.29)	-14.021*** (-4.83)	2.884*** (6.56)	2.680*** (6.52)
sign	-0.910*** (-19.31)	1.375*** (6.57)	-0.873*** (-18.95)	-0.770*** (-18.86)
volatility	0.539*** (12.14)	-1.471*** (-4.61)	0.499*** (11.63)	0.452*** (10.86)
confidence			-0.027*** (-12.81)	-0.024*** (-11.68)
forecast				-0.079*** (-12.49)
Controls	Y	Y	Y	Y
Subject FE	Y	Y	Y	Y
Adj. R^2	0.3165	0.6615	0.3415	0.4133
Obs.	10,040	10,040	10,040	10,040

The main regression is

$$y_{ij} = \beta_0 + \beta_1 \cdot \text{recency}_j + \beta_2 \cdot \text{cluster}_j + \beta_3 \cdot \text{sign}_j + \beta_4 \cdot \text{volatility}_j + \gamma_i + \text{controls} + \varepsilon_{ij} \quad (3.1)$$

where y_{ij} is the perceived risk of subject i for chart j measured by either risk_score_{ij} or confidence_{ij} . On the right-hand side, recency_j , cluster_j , and sign_j are the recency, cluster, and sign factor scores of chart j computed based on definitions in (2.1)-(2.3); volatility_j is the standard deviation of returns of chart j ; γ_i is fixed effect of subject i ; control variables include the order of chart j 's appearance for subject i , subject i 's page time and number of

¹⁹Figure 9 and Figure 10 in Appendix A1 show the price charts with nine highest and lowest average risk scores, respectively. Figure 8 shows the frequency of each price chart.

clicks on chart j , and skewness and kurtosis of returns of chart j .

Column (1) and (2) of Table 2 present results of regression (3.1) with $y_{ij} \equiv \text{risk_score}_{ij}$ and $y_{ij} \equiv \text{confidence}_{ij}$, respectively. Consistent with the findings from Experiment 1, higher recency, more clustering, and more negative sign lead to higher perceived risk and lower forecast confidence. This pattern holds true after controlling for subjective expected returns, showing that the effect does not purely work through return expectations.

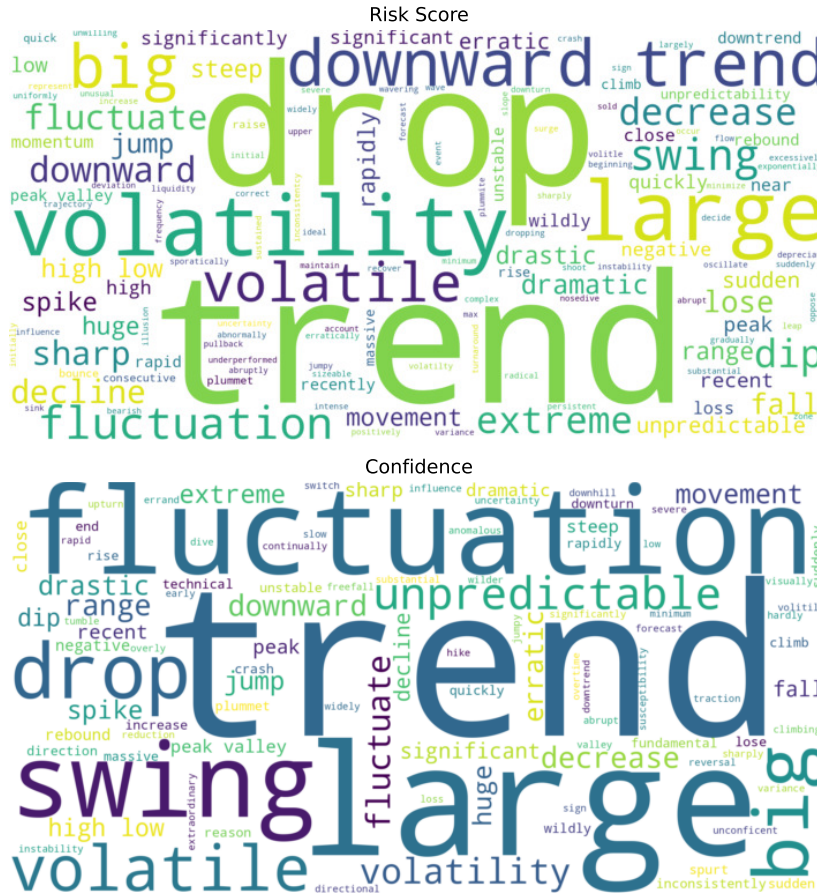
In Appendix A3, we show results from an additional experiment that the effects for the three features are robust to variations in the frequency and range of the historical price data in the charts.²⁰

Self-reported reasoning. In addition to quantitative measures, we also ask participants in open-ended questions what kinds of price charts make them feel risky or unconfident in their price forecasts. Figure 3 provides word clouds for keywords mentioned in participants' responses to these open-ended questions. The keywords mentioned most frequently are similar between answers to two questions, including "downward trend," "volatility," "fluctuation," "large," "spike," "range," "recent," "close," etc. Among these keywords, "downward trend" corresponds to our sign factor; "large," "spike," and "range" correspond to our cluster factor; "recent" and "close" correspond to our recency factor. Many other keywords are also related to our three features. While we do not claim the three features explored in this paper are an exhaustive list of all possible factors, keywords in Figure 3 suggest these three features do appear to be the most relevant ones.

Mechanisms. The effects of recency, clustering, and sign on risk perceptions can work through at least two channels. First, these features could affect risk perceptions by affecting participants' perceptions of volatility. This channel has been confirmed by results from column (2) in Table 2. Second, these features could affect risk perceptions independent from the volatility of stock returns. To test if the second channel plays a role, we include confidence (along with forecast) as a control variable to regression (3.1) with dependent variable being

²⁰Borsboom et al. (2022) shows that shorter time frames of historical price charts increase trading activity but do not affect average risk-taking.

Figure 3: Reasoning for Higher Perceived Risk



remain significant even after controlling for forecast. Taken together, these results indicate that the effects of recency, clustering, and sign on risk perceptions are partially mediated by their impact on perceived volatility. However, a significant portion of these effects occurs independently of perceived volatility, directly influencing risk perceptions.

Willingness to invest. Similar to what we did in Experiment 1, we are interested in the real impact of risk perceptions on investment decisions. The classic risk-return trade-off indicates a negative correlation between investment and risk perceptions, controlling for the expected return. Our hypothesis is that the three features (recency, cluster, and sign) contributing to higher perceived risk should lead to lower investment and that the two risk perception measures (risk score and confidence) should be strongly associated with investment decisions. To formally test this hypothesis, we consider the following regressions

$$\text{inv}_{ij} = \beta_0 + \beta_1 \cdot \text{recency}_j + \beta_2 \cdot \text{cluster}_j + \beta_3 \cdot \text{sign}_j + \beta_4 \cdot \text{volatility}_j + \gamma_i + \text{controls} + \varepsilon_{ij} \quad (3.2)$$

$$\text{inv}_{ij} = \xi_0 + \xi_1 \cdot \text{risk_score}_{ij} + \xi_2 \cdot \text{confidence}_{ij} + \xi_3 \cdot \text{forecast}_{ij} + \xi_4 \cdot \text{volatility}_j + \gamma_i + \text{controls} + e_{ij} \quad (3.3)$$

where inv_{ij} is the investment percentage chosen by subject i for chart j ; and controls are the same as those of regression (3.1). While regression (3.2) uses physical factors which are independent from subject responses as explanatory variables, regression (3.3) focuses on subjective factors reported by subjects.²¹

Column (1) of Table 3 confirms the first part of our hypothesis - higher recency, more same-sign clustered, and more negative sign lead to lower investment. Column (2)-(4) verify the second part of our hypothesis - higher risk perception, measured by both higher risk score and lower confidence, are negatively associated with percentage of investment. It is important to note that with the inclusion of risk score in column (2) and (4), the objective volatility is

²¹Both regression specifications control for returns. (3.2) has the sign factor, which is the objective mean return of the price chart; (3.3) controls for the subjective return by including forecast.

Table 3: Investment

This table reports the estimates from the ordinary least squares (OLS) regression given by (3.2) and (3.3) using the data from Experiment 2. For each price chart j , volatility is the standard deviation of returns; the recency, cluster and sign scores are computed by (2.1), (2.2), and (2.3), respectively. Risk score, forecast, confidence, and investment are specific to and thus indexed by subject i , which are collect from our questions under price chart j . Standard errors are clustered on subject. Robust t -statistics are presented in parentheses. ***, **, and * indicate statistical significance at 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	investment			
recency	−1.759*** (−4.30)			
cluster	−10.092* (−1.93)			
sign	7.414*** (12.86)			
risk score		−5.306*** (−25.07)		−5.044*** (−24.33)
confidence			0.333*** (12.14)	0.191*** (8.44)
forecast		0.904*** (10.86)	1.354*** (13.18)	0.902*** (10.98)
volatility	−3.679*** (−6.43)	−0.497 (−1.06)	−3.452*** (−6.49)	−0.340 (−0.73)
Controls	Y	Y	Y	Y
Subject FE	Y	Y	Y	Y
Adj. R^2	0.4098	0.6141	0.5241	0.6209
Obs.	10,040	10,040	10,040	10,040

no longer significantly affecting investment, which suggests that perceived risk measured by risk score is perhaps a more informative index than the classic objective volatility measure.

To explore the channels through which risk perceptions affect investment decisions, we fit the invested amounts to a modified Merton model (Merton, 1969), where risk scores potentially distort both the numerator (return) and the denominator (risk) of the Merton fraction. In Appendix A2, we report the measure of fit for a non-linear least squares regression with Merton fraction as the functional form. We find that a higher subjective risk score reduces willingness to invest in risky asset through both negatively impacting the numerator and positively impacting the denominator of the Merton fraction.

Explanatory power. While we have shown that the three features significantly affect risk perceptions, it remains to be shown how the three-feature model compares with other models

in terms of their explanatory power. To study this, we evaluate the in-sample R^2 of a linear regression model including these features and compare it to several benchmarks. Specifically, we use the Experiment 2 dataset with observations for real stock price charts and calculate the R^2 of 6 models, each regressing the risk scores on a unique set of independent variables. The included independent variables and the values of R^2 of these models are shown in Table 4. The results show that our three-feature model (Model 3) more than doubles the explanatory power of a model that only include the second, third, and fourth moments of the return distributions. Moreover, our model has a R^2 that is close to those of existing models that include far more explanatory variables, and it explains 80% of the explainable variations in the data, measured by the R^2 of a model with chart fixed effect (Fudenberg et al., 2022).

Table 4: Explanatory Power of Six Models

This table reports the R^2 of 6 different regression models. All regressions have the same dependent variable as the risk score but different independent variables. From 1 to 6, the model becomes progressively more sophisticated. Our proposed 3-feature model is Model 3. The data used for all 6 regressions are from the real stock price charts of Experiment 2.

Model	Variables	R^2
1	Objective standard deviation of each price chart	4.1%
2	Model 1 plus skewness, and kurtosis	4.2%
3	Model 2 plus recency, sign, and cluster	10.1%
4	Model 2 plus all variables considered in Borsboom and Zeisberger (2020) ²²	11.1%
5	Model 3 plus all variables considered in Borsboom and Zeisberger (2020)	11.3%
6	Graph fixed effects	13.8%

Predictive power. To assess the predictive power of the three proposed features, we evaluate their out-of-sample predictions using the Experiment 2 dataset with observations for real stock price charts. Specifically, we randomly split the dataset into a 80% training set and a 20% testing set. Within the training set, we estimate ordinary least squares (OLS) coefficients of Models 1 to 5. We then apply the fitted OLS models to the testing set and calculate the out-of-sample R^2 . We repeat this procedure 100 times, each with a different randomly

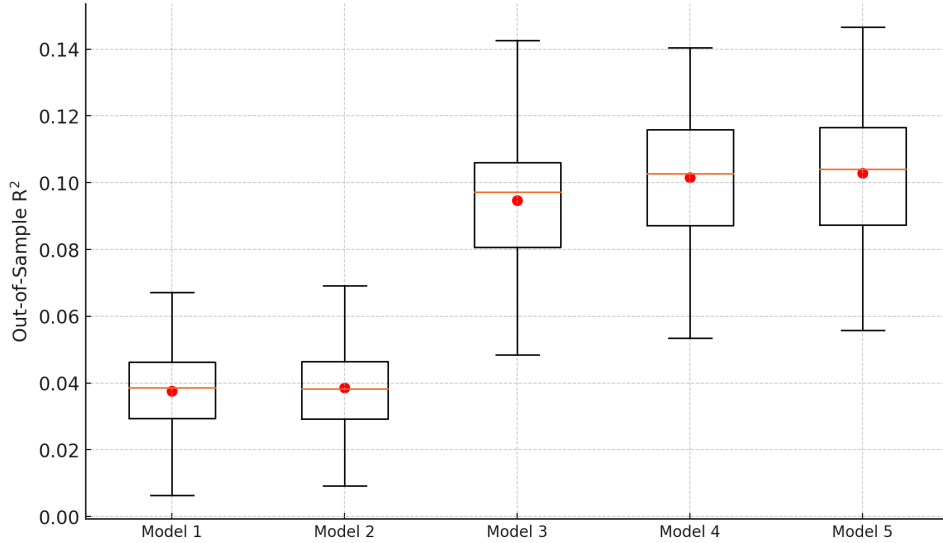
²²These variables include: maximum price within 12 months, minimum price within 12 months, maximum price decrease within a month, maximum price increase within a month, price change in the last 3 months, monthly standard deviation of second half minus monthly standard deviation of first half, total return of second half minus total return of first half, number of months the chart moves below the initial price, and average return.

generated training-testing split, yielding 100 out-of-sample R^2 values for each model. The distributions of these R^2 values are depicted in Figure 4.

Among the five models, the largest improvement in predictive power occurs when we add our proposed features to the three moments of the price charts, increasing out-of-sample R^2 from an average of 3.5% to 9.5%. The marginal improvement from replacing our three features with or further adding the features considered in [Borsboom and Zeisberger \(2020\)](#) is much smaller, and the distributions of R^2 show considerable overlaps, despite the large number of features included.

Figure 4: Model Comparison with Out-of-Sample R^2

This figure is a box-and-whisker plot of the distributions of out-of-sample R^2 of Model 1-5 listed in Table 4. The box represents the inter-quartile range and the whiskers represent the range. The line and the dot in the box represent the median and the mean, respectively.



Taken together, these results indicate that the three features we propose significantly improve the explanation and prediction of risk perceptions. Moreover, the performance of the three-feature model is close to those of much more complex models despite its parsimony and portability.

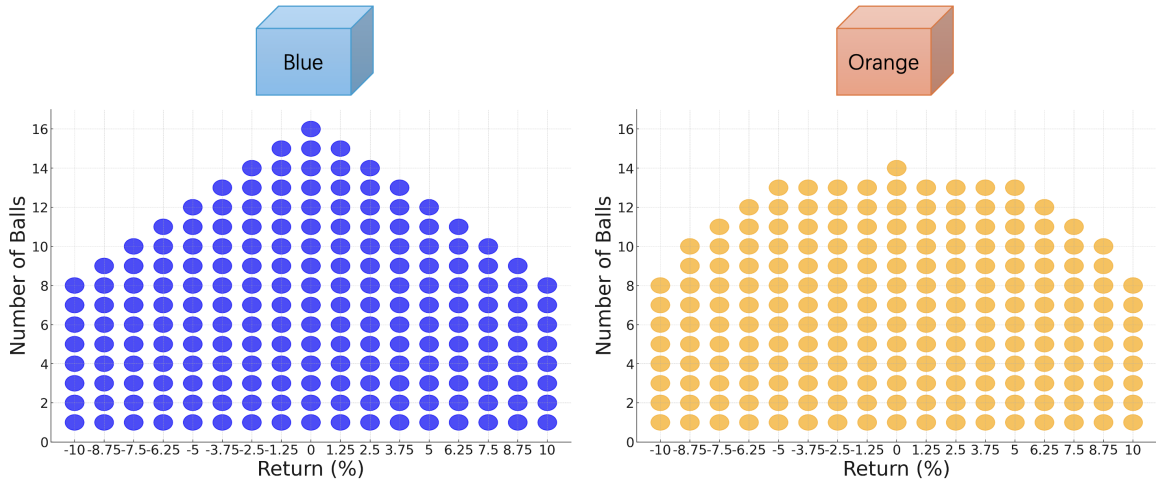
4 Experiment 3

4.1 Design

The mechanism analysis for Experiment 2 data has established that our three proposed features can affect risk perceptions both through their effects on perceived volatility and independent of volatility. While risk perceptions, as measured by risk scores, are a purely subjective construct, perceived volatility as measured by participants' confidence in their forecasts is a belief variable that has objective benchmarks. To investigate whether the effects of our three features on perceived volatility are consistent with Bayesian belief updating, we conduct a third experiment where the data-generating process is fully specified and, hence, volatility perceptions have a clear Bayesian benchmark.

Figure 5: Two Possible Distributions of Returns for the Experiment 3

The figure illustrates instructions received subjects on how returns in Experiment 3 are determined. Blue box on the left-hand side and orange box on the right-hand side each corresponds to a distribution of returns. Subjects see this diagram in Experiment 3 and are told that returns are determined by the numbers on the balls that are randomly drawn from one of these two boxes, and that the draws for the same price chart will be conducted from the same box.



Specifically, in Experiment 3, we ask participants to consider an asset whose returns are independently and identically distributed (i.i.d.) across months. The distribution is equally likely to be determined by the blue box and orange box shown in Figure 5, with a

realization of return represented by a ball randomly drawn from the box. Given these two possible distributions, we ask participants to rate the riskiness of an asset assuming its price follows the path depicted in a price chart. We also elicit the posterior beliefs about the true return distribution, incentivized to be close to the Bayesian answer, which is our measure of perceived return volatility. Participants are presented with six charts from Experiment 1 in random order: Benchmark, Early, Recent, Clustered, Positive, and Negative. Because the returns of the first four charts are permutations of each other, the Bayesian posterior beliefs are the same given that returns are i.i.d across months. For the Positive chart, we inform participants that the distributions is shifted upwards by 1.125 percentage points. For the Negative chart, the distribution is shifted downwards by the same amount. 80 participants from Prolific participated in this experiment and the median duration was 10 minutes. Everyone is paid a participation fee of \$3 with a chance of an additional \$3 bonus.²³

4.2 Result

Column (1) of Table 5 shows the OLS estimates of regressing posterior beliefs about the return-generating process on recency, clustering, and sign using data from Experiment 3.²⁴ Even in this setting where the Bayesian posteriors should not depend on these three features, participants’ beliefs are strongly influenced by them in the same directions as in Experiments 1 and 2. This result suggests that the effects of the three features on perceived volatility of the return-generating process are not fully explained by Bayesian updating. It also indicates that the effects persist when participants have full information about the assets’ return-generating processes.

In Column (2), we replace the dependent variable with the risk scores. Clustering and sign continue to have an effect on risk perceptions, while the effect of recency is no longer significant.²⁵

²³Details of Experiment 3 were pre-registered and are available at https://aspredicted.org/82V_JN3.

²⁴We omit the standard deviations of the price charts from this regression because all price charts presented in Experiment 3 have the same return standard deviation.

²⁵One potential reason for the muted effect of recency on risk perceptions could be the gambler’s fallacy.

In Column (3), we compare the effects of the three features on risk scores between Experiments 1 and 3, taking advantage of the fact that the six price charts presented in Experiment 3 are also used in Experiment 1. This comparison allows us to explore whether the effects differ between a setting with fully-specified data-generating processes and another where participants may bring in their subjective beliefs about real stocks. The coefficients on the interaction terms between the features and the Experiment 3 indicator variable show that the effects of recency and clustering are dampened in Experiment 3, while the effect of sign is not significantly different. These results suggest that home-grown beliefs about real stocks may play a role in the effects of recency and clustering on risk perceptions.

Table 5: Comparison of Experiment 1 and 3

This table reports the estimates from the ordinary least squares (OLS) regression given by (3.1) using the data from both Experiment 1 and 3. For each price chart j , std is the standard deviation of returns; the recency, cluster and sign scores are computed by (2.1), (2.2), and (2.3), respectively. Standard errors are clustered on subject. Robust t -statistics are presented in parentheses. ***, **, and * indicate statistical significance at 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)
	posterior	risk score	
	Experiment 3	Experiment 3	Experiments 1 & 3 Pooled
recency $\times$ Experiment 3			-0.447*** (-4.12)
cluster $\times$ Experiment 3			-4.360*** (-2.90)
sign $\times$ Experiment 3			-0.067 (-0.32)
recency	2.128** (2.16)	-0.085 (-0.96)	0.352*** (5.39)
cluster	91.418*** (6.78)	5.965*** (5.09)	8.757*** (9.52)
sign	-5.804*** (-3.82)	-1.308*** (-7.69)	-1.206*** (-9.36)
std			0.330*** (10.13)
Controls	Y	Y	Y
Subject FE	Y	Y	Y
Adj. R^2	0.2538	0.2838	0.3051
Obs.	480	480	1,280

Specifically, after a recent period of large fluctuations, participants who believe in the gambler's fallacy may expect a subsequent period of small fluctuations, even though they believe the data-generating process for returns has high variance.

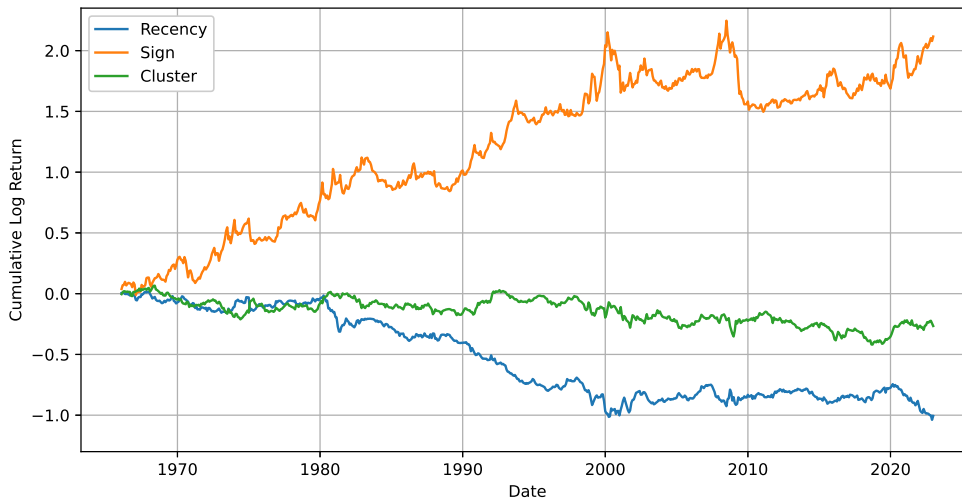
5 Applications

While we have shown that recency, clustering, and sign significantly and robustly affect risk perceptions and investment decisions in an experimental setting, one may still wonder whether these effects are relevant in real-world financial markets. In this part, we extend what we have found so far to real investment scenarios with observational data on two levels: individual stocks and mutual funds. One disadvantage of risk perception measures is that they are subjective and typically unobserved outside of experimental settings. Our approach to dealing with this challenge is to focus on the three measures we have explored, i.e., recency, clustering, and sign, to unveil their asset pricing implications, which in turn reveals whether investors care about these dimensions in their decision-making process.

5.1 Individual Stock

Figure 6: Cumulative Log Returns

The figure shows the cumulative log returns of high-minus-low portfolios for recency, cluster and sign. The portfolios are constructed with CRSP monthly returns from 1966 to 2022. For each stock in month t , the recency score, sign score, cluster score, and return standard deviation are computed using returns from month $t - 12$ to $t - 1$. Then, stocks are sorted based on each of the three feature scores within each volatility decile to form high (top 30 percentile) and low (bottom 30 percentile) portfolios for that feature in month t . Logarithmic value weighted returns of high and low in all deciles are used to compute volatility adjusted period feature-return difference as an average across ten deciles, and finally the period feature returns are summed across all past periods to generate the cumulative returns shown on the figure.



One implication is about individual stock returns. Figure 6 presents the cumulative log returns of high-minus-low portfolios for each of the three features from 1966 to 2022.²⁶ To construct these portfolios, we use the CRSP monthly returns and follow the standard data processing procedures.²⁷ For each stock in month t , we first compute the recency score, sign score, cluster score, and return standard deviation using returns from month $t - 12$ to $t - 1$. We then sort stocks based on their return volatility into ten deciles and construct high (top 30 percentile) and low (bottom 30 percentile) portfolios for recency, sign, and cluster within each volatility decile to obtain value-weighted returns, i.e., $vw_ret_{i,j,d,t}$ for $i \in \{\text{high}, \text{low}\}$, $j \in \{\text{recency}, \text{sign}, \text{cluster}\}$, and $d \in \{1, \dots, 10\}$. We finally compute the volatility-adjusted log return for feature j in month t by

$$ret_{j,t} = \frac{1}{10} \sum_{d=1}^{10} [\log(1 + vw_ret_{\text{high},j,d,t}) - \log(1 + vw_ret_{\text{low},j,d,t})] \quad (5.1)$$

and the cumulative log return by

$$cml_ret_{j,t} = \sum_{\tau \leq t} ret_{j,\tau}. \quad (5.2)$$

As given by equation (5.1)-(5.2), the volatility-adjusted cumulative return $cml_ret_{j,t}$ controls for volatility and highlights the effect of each of the three features we have explored in our experiment. For a stock with a higher perceived risk, the market reacts differently in the short run as opposed to the long run. In the short run, market may react negatively to a less favorable stock (i.e., the one with a higher perceived risk), which leads to an immediate lower return but a follow up higher expected return in the long run to compensate for the higher risk. We focus on the next month short-term relative returns to avoid complications

²⁶We impose the restriction of having at least 30 stocks in each side of the constructed portfolios, and the year of 1966 is the earliest year such that this restriction is satisfied in all years afterwards.

²⁷We include all stocks with share codes of 10 and 11 and exchange codes of 1, 2, and 3. We adjust for delisting returns according to [Shumway \(1997\)](#) to avoid bias in CRSP returns. We keep the observations with non-zero price, positive volume, and returns not equal to -66, -77, -88, or -99. We then drop penny shares that are traded below \$5 and micro-cap stocks whose market capitalization at the end of last December fall into the bottom decile.

from reversal and persistence of these features in later months.

Figure 6 shows that higher recency score, lower sign score, and higher cluster score, which all correspond to higher perceived risk in our experiments, are associated with lower returns in the immediate next month. While we do not claim there is any causality in this relation, our conjecture is that stocks with higher perceived risk incurred through our three features leads to lower willingness of investment, which is then reflected in their relatively worse market performance in the short-run. In the Appendix A4, we present the graph with sorting on shocks to the three features (Figure 13) and the graphs with longer lags of 3, 6, 9, and 12 months (Figure 14) between the last month of the period for calculating three feature scores and the month of portfolio rebalancing. These additional tests suggest that the recency feature tend to be persistent, while cluster and sign features are relatively transitory with return reversals later.

With the dataset of real stocks, we can compute the correlations between these features. Table 6 presents the cross-sectional average of the time-series correlations between any pair from recency, cluster, sign, and volatility of every stock. It is clear that all three features we highlight in this paper are relatively independent from each other and from volatility, which further corroborates our selection of these three features for parsimony.

Table 6: Correlations

This table reports the average correlation between each pair formed from recency, cluster, sign, and volatility using monthly CRSP data between 1966-2022, which contains 10,784 unique permnos and 1,252,140 firm-by-month observations.

	recency	cluster	sign	volatility
recency	1.000			
cluster	0.018	1.000		
sign	-0.068	-0.125	1.000	
volatility	0.017	-0.072	0.096	1.000

In addition to returns, we also look into the relations between our three features and other variables, including trading volume and future volatility. We run the following regression

using the same monthly market data as shown in Figure 6:

$$y_{jt} = \beta_1 \cdot \text{recency}_{jt} + \beta_2 \cdot \text{cluster}_{jt} + \beta_3 \cdot \text{sign}_{jt} + \beta_4 \cdot \text{volatility}_{jt} + \gamma_j + \gamma_t + \text{controls} + \varepsilon_{jt}. \quad (5.3)$$

In Table 7, y_{jt} represents the trading volume of stock j in month t in column (1)-(2) and the future 12-month (i.e., t to $t + 11$) volatility of stock j in column (3)-(4). recency_{jt} , cluster_{jt} , and sign_{jt} are the three feature scores computed using past 12-month (i.e., $t - 12$ to $t - 1$) data. γ_j and γ_t are the stock and time fixed effects, respectively.

Table 7: Market Data

This table reports the estimates from the ordinary least squares (OLS) regression given by (5.3) using monthly CRSP data between 1966-2022. For each stock j in month t , volatility is the standard deviation of its returns from month $t - 12$ to $t - 1$; the recency, cluster and sign scores are computed by (2.1), (2.2), and (2.3), respectively, using the same set of past returns. Volume is the dollar-amount volume of the stock in month t . Future volatility is the standard deviation of the returns from month t to $t + 11$. Standard errors are clustered on permno and month. Robust t -statistics are presented in parentheses. ***, **, and * indicate statistical significance at 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	Volume		Future Volatility	
recency	0.002*** (3.76)	0.003*** (13.79)	0.319*** (7.47)	0.164*** (14.60)
cluster	0.040*** (9.04)	0.006** (2.40)	2.302*** (9.92)	0.581*** (4.28)
sign	-0.005*** (-16.95)	-0.006*** (-23.33)	-0.082*** (-5.43)	-0.034*** (-3.23)
Controls	1	Full	1	Full
FE	N	Y	N	Y
Adj. R^2	0.1645	0.4625	0.3137	0.5747
Obs.	1,397,748	1,397,748	1,264,824	1,264,824

As shown in Table 7, we find that higher recency score, higher cluster score, and lower sign score, which again represent higher perceived risk based on our experimental results, are all associated with higher trading volume and higher future volatility. Although we cannot make causal claim on these results, we do find them plausible and consistent with our experimental results in the sense that higher perceived risk corresponds more uncertainty and disagreement in expectations, which then leads to more trading. In addition, it seems

that investors are correct in identifying riskier stocks as what feels more risky, and those stocks indeed turn out to have higher volatility going forward even if they currently have the same volatility.

5.2 Mutual Fund

We then turn to the mutual fund data to check if flows are driven by our three features. This is a useful exercise as mutual fund investors receive fund prospectus periodically, which always include a performance chart of the fund recently. Investors can also login to the fund management company’s website to see the price chart of their funds at different horizons. So, it is natural that our lab findings may extend to the real mutual fund investment scenarios.

We obtain the monthly total net assets and returns data from WRDS for 10,809 mutual funds from 2000 to 2023. We adopt the classic approach to obtain fund flows on the monthly frequency (Coval and Stafford, 2007). Specifically, the dollar amount flow, $FLOW_{j,t}$, of fund j in month t is defined by

$$FLOW_{j,t} = TNA_{j,t} - TNA_{j,t-1} (1 + r_{j,t})$$

where $TNA_{j,t}$ and $r_{j,t}$ are the total net assets and return on the investment of fund j in month t . Then, following Coval and Stafford (2007), we define the percentage flow as follows,

$$flow_{j,t} = \frac{FLOW_{j,t}}{TNA_{j,t-1}},$$

which measures the magnitude of change in investor demand and is homogeneous of degree 0 with respect to the fund size.

Since $flow_{j,t}$ is a variable of change, we focus on the effect of the changes in our three feature scores computed using past three year’s monthly net asset value per share for each fund in our sample. To reduce the noise in our dependent variable, we construct negative flow indicators $\mathbb{I}\{x\% \text{ Out Annually}\}$, which equals to 1 if the cumulative percentage flow in

the past year (half-year) reaches $x\%$ ($\frac{x}{2}\%$) in the annual (semi-annual) regression. We also roll over the feature score determination sample by 1 year (6 months) at a time for each fund to ensure significant price chart updating.

Table 8: Mutual Fund Negative Flows

This table reports the estimates from the ordinary least squares (OLS) regression given by (5.4) using monthly WRDS mutual fund data between 2000-2023. For each fund j in month t , $\Delta\text{volatility}$ is the change in standard deviation of its returns from $[t - 36 - k, t - 1 - k]$ to $[t - 36, t - 1]$, where the k is 6 (12) for the semi-annual (annual) regression. The recency, cluster and sign scores are computed by (2.1), (2.2), and (2.3), respectively, and the changes in those scores are computed using the same method as $\Delta\text{volatility}$. The control variables include the change in skewness and kurtosis computed in the same way. Standard errors are clustered on Cusip. Robust t -statistics are presented in parentheses. ***, **, and * indicate statistical significance at 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	$\mathbb{I}\{10\% \text{ Out Annually}\}$		$\mathbb{I}\{20\% \text{ Out Annually}\}$		$\mathbb{I}\{30\% \text{ Out Annually}\}$	
	Semi-Annual	Annual	Semi-Annual	Annual	Semi-Annual	Annual
$\Delta\text{recency}$	0.020*** (10.55)	0.007*** (3.09)	0.017*** (9.89)	0.005** (2.42)	0.013*** (9.09)	0.005*** (3.27)
$\Delta\text{cluster}$	0.169*** (3.63)	0.215*** (4.62)	0.097** (2.41)	0.232*** (5.98)	0.062* (1.87)	0.148*** (4.92)
Δsign	-0.055*** (-27.97)	-0.044*** (-23.52)	-0.041*** (-23.77)	-0.032*** (-20.39)	-0.029*** (-20.12)	-0.019*** (-15.51)
$\Delta\text{volatility}$	0.010*** (23.73)	0.009*** (21.72)	0.008*** (19.56)	0.007*** (17.58)	0.006*** (15.84)	0.004*** (13.79)
Controls	Y	Y	Y	Y	Y	Y
Cusip FE	Y	Y	Y	Y	Y	Y
Adj. R^2	0.1414	0.1429	0.1197	0.1169	0.0965	0.0870
Obs.	220,911	112,692	220,911	112,692	220,911	112,692

Our experimental results indicate that a higher perceived risk, which corresponds to positive recency and cluster score change and negative sign score change, leads to lower willingness to invest, translating into higher likelihood of triggering a significant negative flow event. To formally test this hypothesis, we run the following regression

$$\begin{aligned} \mathbb{I}_{j,t}\{x\% \text{ Out Annually}\} = & \psi_1 \cdot \Delta\text{recency}_{j,t} + \psi_2 \cdot \Delta\text{cluster}_{j,t} + \psi_3 \cdot \Delta\text{sign}_{j,t} + \psi_4 \cdot \Delta\text{volatility}_{j,t} \\ & + \gamma_i + \text{controls} + \varepsilon_{j,t}, \end{aligned} \quad (5.4)$$

of which the estimates are reported in Table 8.

To show robustness, we present results for both semi-annual and annual rolling windows across three different level of flow out indicator thresholds, i.e., 10%, 20%, and 30%. The results are all showing that higher risk perception leads to lower willingness to invest.

The fact that we see this consistent pattern on both individual stock level and mutual fund level suggest that the effect of risk perceptions on willingness to invest is not restricted to lab settings; instead, it leads to actual movement of money by investors in real investment scenarios.

6 Conclusion

Our experiments demonstrate that three conceptually unique and parsimonious features—recency, clustering, and sign—significantly influence investors’ perceptions of risk. We find that price charts with higher recency and clustering, as well as negative sign in return series, are perceived as riskier. These perceptions translate into reduced willingness to invest, highlighting the real impact these features exert on investment behaviors. On the mechanisms of these effects, we show that they operate both through their effects on perceived volatility and independent of volatility. Moreover, the former effect is not fully explained by Bayesian inferences about the return data-generating process.

We further test the implications of our experimental findings by analyzing real financial market data. We find that stocks that are perceived riskier through our three features earn lower return and trade in higher volume in the next month, and have higher future 12-month volatility. We also compute the three features’ scores for U.S. mutual funds and find that funds with greater increase in perceived risk level with our three features are more likely to experience a significant negative flow.

Our paper builds on the existing literature on risk perception and paves the way for future studies to further investigate the parsimonious set of features that systematically shape risk perceptions and investment decision-making. On a broader picture, the wedge between any

“physical” measure in the financial market and its “perceived” counterpart is worth studying. More work shall be done to bridge the experimental findings about such wedges to real data applications.

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Appendix A1: Additional Figures for Experiment 1-3

Figure 7: All Price Charts in Experiment 1

The figure shows all the price charts used in Experiment 1. The first 7 are fabricated price charts obtained by reshuffling the same set of returns, while the last one is a real price chart with standardized price levels.

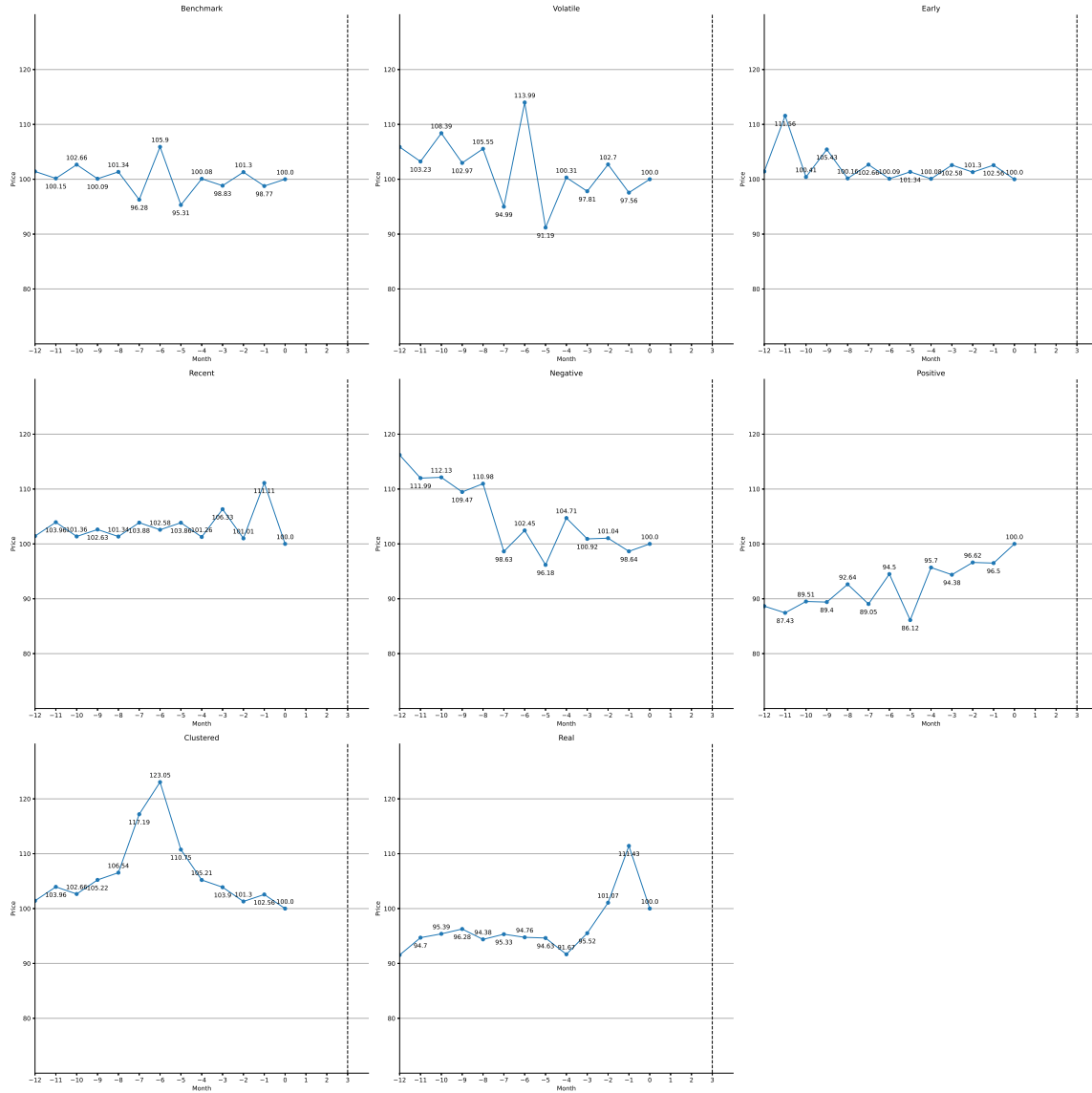


Figure 8: Number of Responses of 200 Figures in Experiment 2

The figure shows the number of responses collected for each of the 100 price charts in the Reshuffle group and each of the 100 price charts in the Real group. All charts in the Reshuffle group are obtained by randomly reshuffling the same set of returns used in Experiment 1. Charts in Real group are obtained by random selection.

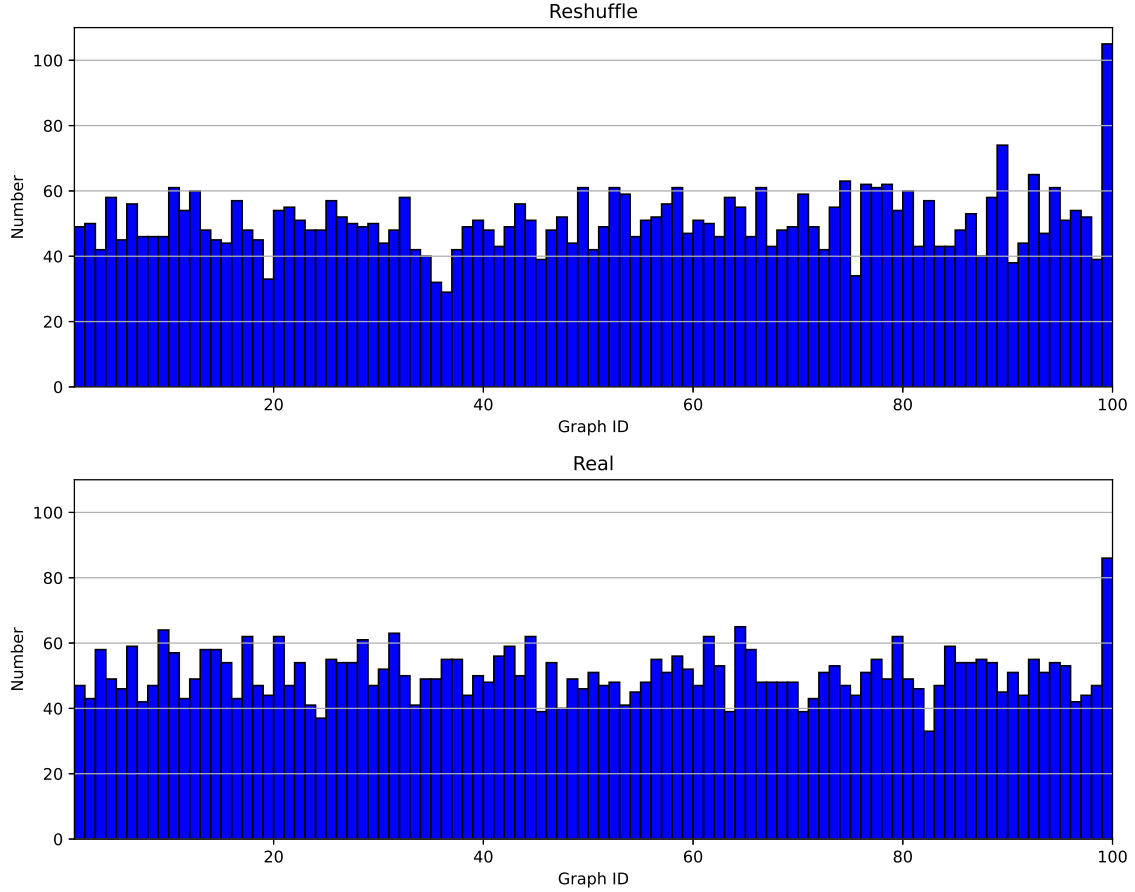


Figure 9: Top Ranked Risk Score Graphs in Experiment 2

The figure shows the top 9 out of 200 price charts with the highest average risk score assessed by subjects in Experiment 2.

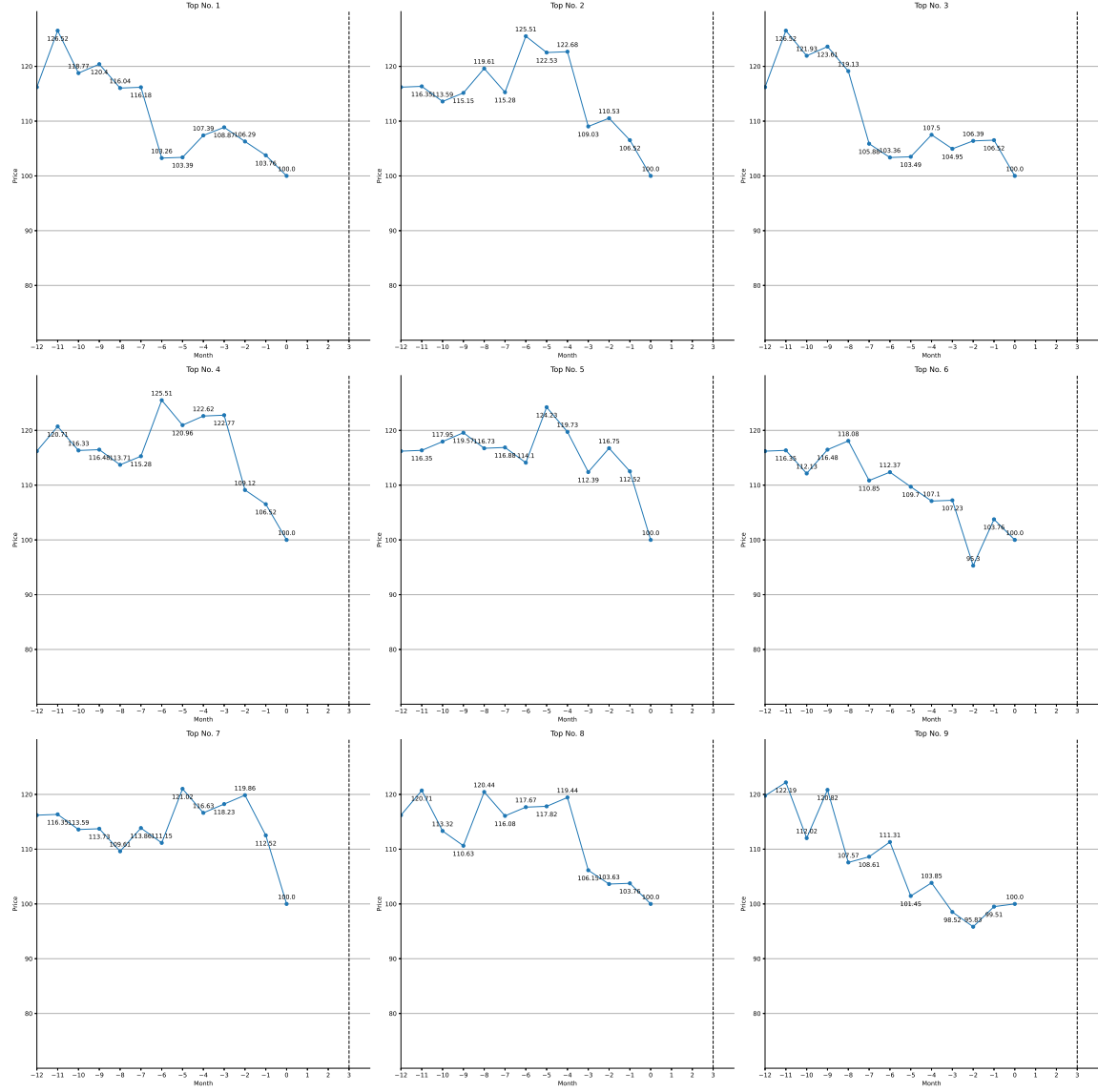
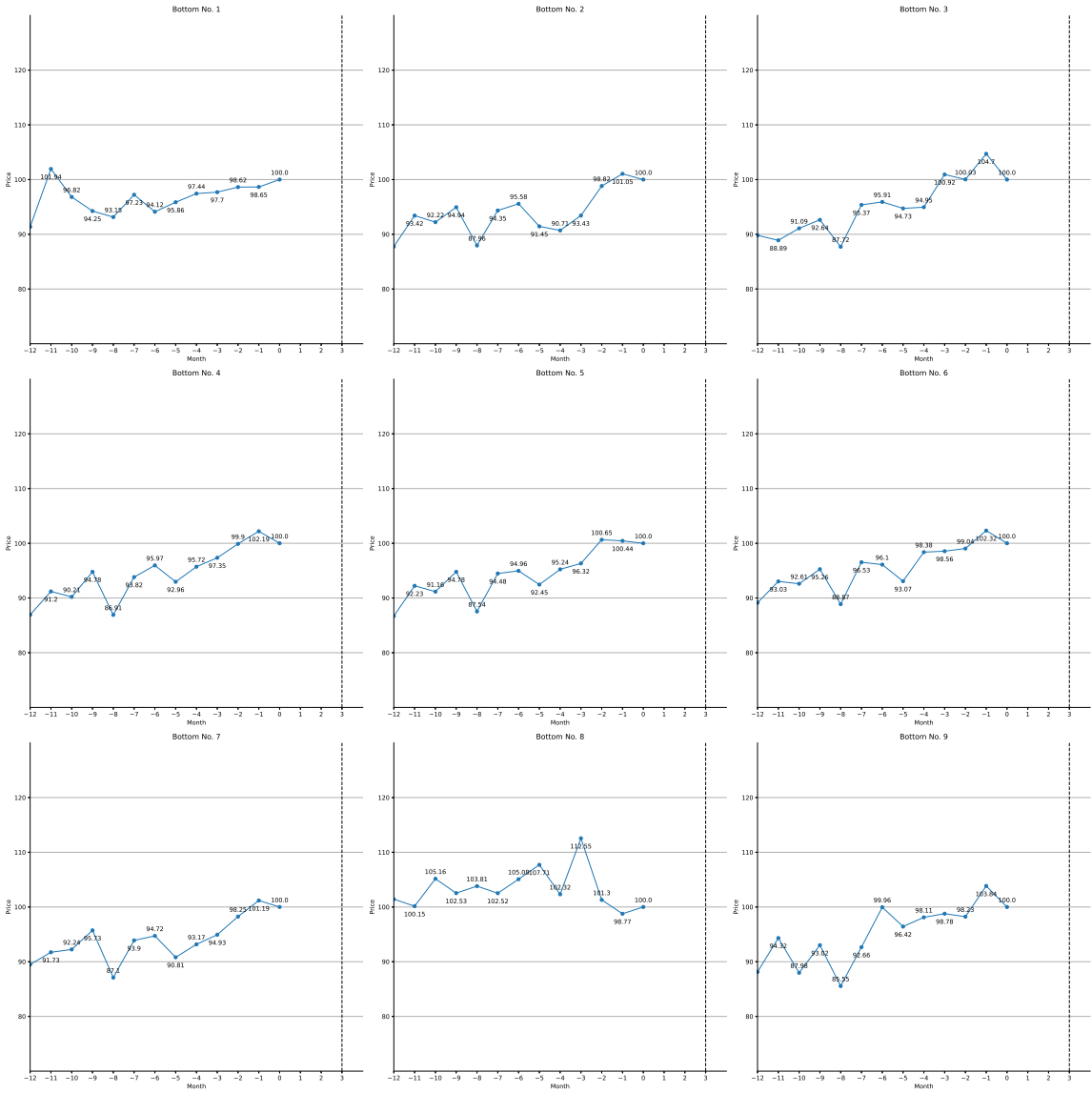


Figure 10: Bottom Ranked Risk Score Graphs in Experiment 2

The figure shows the bottom 9 out of 200 price charts with the lowest average risk score assessed by subjects in Experiment 2.



Appendix A2: Non-linear Least Squares Regression with Merton Fraction

We look into how risk score affects investment decision making. Note that the investment task in our experiment resembles Merton's problem [Merton \(1969\)](#). Therefore, we start with the benchmark of Merton's fraction with zero risk-free rate, i.e., $f = \mu/\gamma\sigma^2$, where μ is the expected risky return, γ is the risk aversion, and σ is the standard deviation of the risky return. We plug in the objective average return and return standard deviation for each price chart to be the μ and σ , and we propose a modified Merton's fraction to be

$$f_{ij}(\xi_\mu, \xi_\sigma) = \frac{\mu_j + \xi_\mu \text{risk_score}_{ij}}{(\sigma_j + \xi_\sigma \text{risk_score}_{ij})^2} \quad (6.1)$$

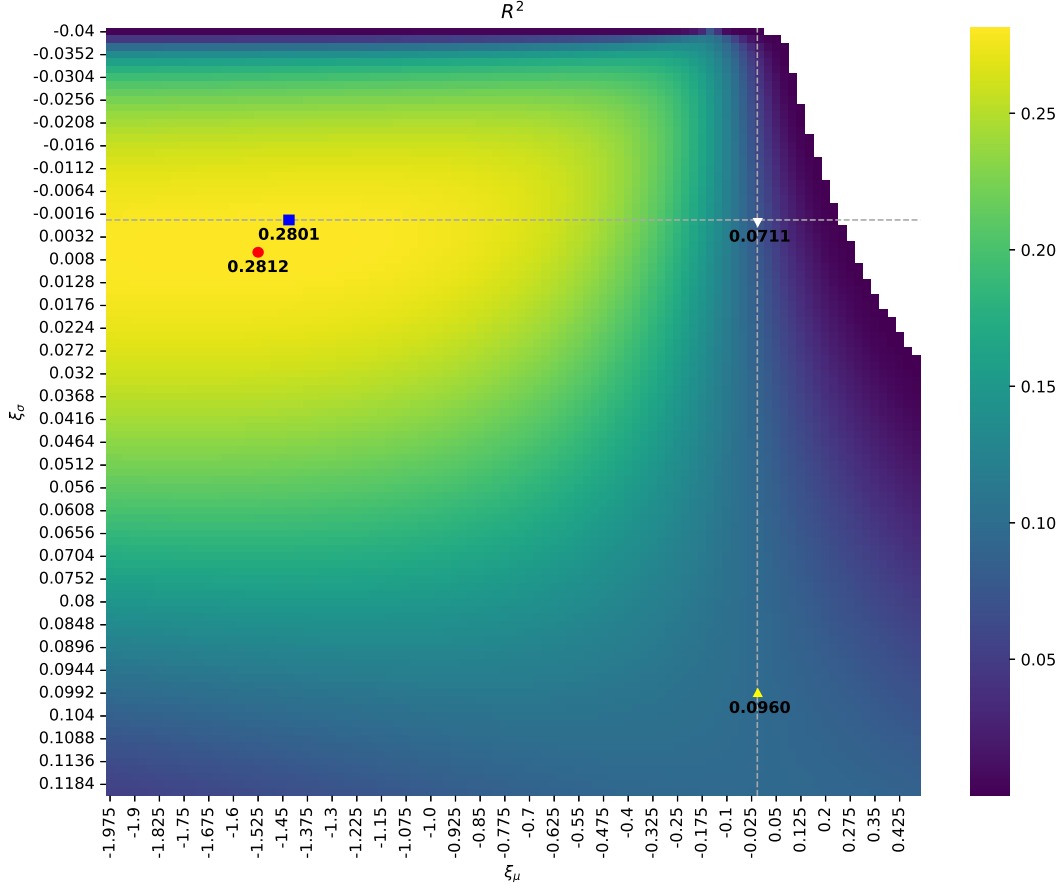
where ξ_μ and ξ_σ govern how investor i 's risk score towards chart j may affect the effective μ_j and σ_j of chart j , respectively. The risk aversion γ is an unobserved positive constant. Since we are interested in the within-subject effect of risk score on the explanatory power of the Merton fraction, γ is absorbed into the regression coefficient of $f_{ij}(\xi_\mu, \xi_\sigma)$. Then, we consider different combinations of values for (ξ_μ, ξ_σ) to obtain the corresponding modified Merton fraction, which will be used in the following regression

$$\text{inv}_{ij} = \beta_0 + \beta_1 \cdot f_{ij}(\xi_\mu, \xi_\sigma) + \kappa_i + \varepsilon_{ij} \quad (6.2)$$

where inv_{ij} is the investment percentage chosen by subject i for chart j ; $f_{ij}(\xi_\mu, \xi_\sigma)$ is the modified Merton's fraction given by (6.1); and κ_i is the subject fixed effect. We estimate the restricted non-linear regression (6.2) for all pairs of values of (ξ_μ, ξ_σ) in reasonable ranges to find the optimal value that generates the highest R^2 .

Figure 11: Optimal Coefficients for Merton Fraction Modified with Risk Score

The figure shows the within R^2 of regression (6.2) for $\xi_\sigma \in [-0.04, 0.12]$ and $\xi_\mu \in [-2, 0.5]$, which contains the global maximum after we test much broader ranges. The horizontal and vertical grey dashed lines stand for $\xi_\sigma = 0$ and $\xi_\mu = 0$, respectively. The red circle marks the coordinates of global maximum; the blue rectangle marks the coordinates of restricted maximum with $\xi_\sigma = 0$; the yellow triangle marks the coordinates of restricted maximum with $\xi_\mu = 0$; and the white inverted triangle marks the coordinates with $\xi_\sigma = 0$ and $\xi_\mu = 0$. The corresponding within R^2 value is reported in black bold number below each marked point.



In Figure 11, we report all the corresponding within R^2 values for those pairs such that the estimated β_1 is positive. The global maximum within R^2 is obtained at $(\xi_\mu^* = -1.55, \xi_\sigma^* = 0.0064)$. Comparing the R^2 at $(\xi_\mu^*, \xi_\sigma^*)$ against that from benchmark model at $(0, 0)$, the model fit increased dramatically from 0.0711 to 0.2812, which indicates that the risk score we have elicited is a useful measure that can significantly improve the explanatory power of standard Merton's fraction in investment decisions when incorporated properly.

To further understand the way risk score gets involved, we compare the global maximum R^2 against the restricted maximum R^2 with $\xi_\mu = 0$ and $\xi_\sigma = 0$, respectively. We find that while the improvement from incorporating risk score is small when setting $\xi_\mu = 0$ (from $R^2 = 0.0711$ to $R^2 = 0.0906$), it is sufficiently large even though setting $\xi_\sigma = 0$ (from $R^2 = 0.0711$ to $R^2 = 0.2801$). This observation suggests that it is likely risk score affects investment decisions mainly by operating through the numerator rather than the denominator of the Merton's fraction. The negative ξ_μ^* and slightly positive ξ_σ^* indicate that a high subjective risk score largely reduces the effective return (numerator) and increases the effective standard deviation (denominator), both of which reduce the willingness to invest in risky asset.

Appendix A3: Alternative Horizon and Frequency

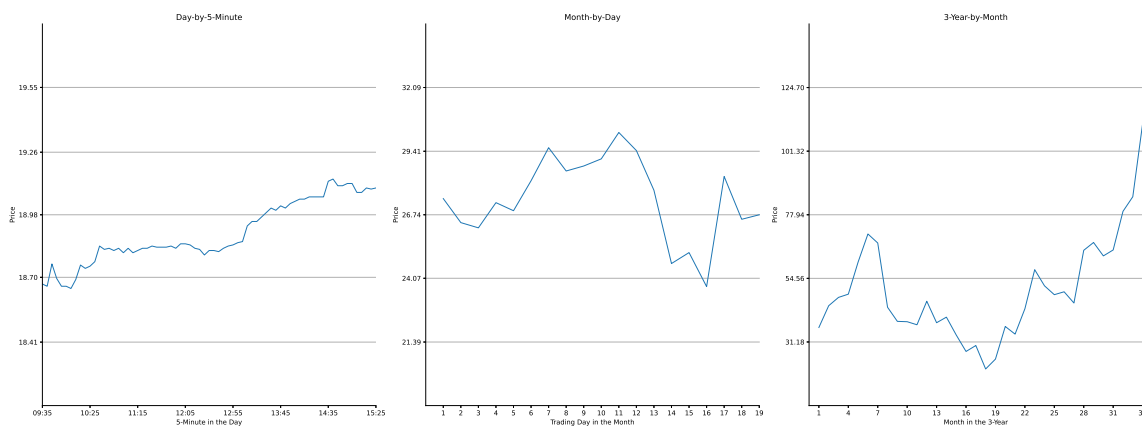
In the three experiments we have analyzed in Section 2-4, the price charts show stock prices by month over a 12-month period. A remaining question is whether the price chart features identified from these charts maintain their explanatory power in charts with alternative frequencies and time frames.

Design

To answer this question, we conduct Experiment 4 ($N_{\text{graph}}^{\text{step } 4} = 90$ and $N_{\text{subject}}^{\text{step } 4} = 90$) where participants rate the riskiness of real companies' stock price charts in three different formats: 3-years-by-month, 1-month-by-day, and 1-day-by-5-minutes (see Figure 12 for examples). Each participant is presented with 10 charts in each format, displayed in random order. The median duration is 10 minutes, and the participation fee is \$4.²⁸

Figure 12: Sample Charts for Experiment 4

The figure shows examples of the price charts in Experiment 4 with different horizon-frequency combinations. For left to right, the chart corresponds to Day-by-5-Minute, Month-by-Day, and 3-Years-by-Month, respectively.



²⁸Details of Experiment 4 were pre-registered and are available at <https://aspredicted.org/V4Y-CMN>.

Result

Following the methodology of Experiment 2, we applied regression (3.1) to the data collected across various horizon-frequency combinations. The detailed results are displayed in Table 9.

While the influence of recency and cluster tends to less stable in regressions with shorter horizons, the overarching patterns remain consistent with those observed in our primary 1-year-by-month analyses. These findings suggest that the impact of recency, cluster, and sign features is not confined to any single horizon-frequency combination but extends broadly across diverse price chart configurations. Therefore, we believe the three features explored in our study represent a set of prevalent biases when investors receive and process information from price charts.

Table 9: Alternative Horizon-Frequency

This table reports the estimates from the ordinary least squares (OLS) regression given by (3.1) using the data from Experiment 4. For each price chart j , volatility is the standard deviation of returns; the recency, cluster and sign scores are computed by (2.1), (2.2), and (2.3), respectively. Risk score is specific to and thus indexed by subject i , which is collect from our question under price chart j . Column 1 to 3 correspond to results under subsamples of Day-by-5-Minute, Month-by-Day, and 3-Years-by-Month, respectively. Standard errors are clustered on subject. Robust t -statistics are presented in parentheses. ***, **, and * indicate statistical significance at 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)
	risk score		
	1-day	1-month	3-year
recency	0.084 (0.40)	0.478*** (4.36)	0.910*** (5.35)
cluster	12.532*** (6.15)	-0.482 (-0.31)	3.189* (1.75)
sign	-32.398*** (-8.57)	-1.613*** (-11.57)	-0.527*** (-5.49)
volatility	2.131** (2.20)	0.407*** (10.97)	0.241*** (9.03)
Controls	Y	Y	Y
Subject FE	Y	Y	Y
Adj. R^2	0.520	0.446	0.327
Obs.	900	900	900

Appendix A4: Additional Figures for Applications

Figure 13: Cumulative Log Returns - Sorted on Shocks to Three Features

The figure shows the cumulative log returns of high-minus-low portfolios for shocks to recency, cluster and sign. The portfolios are constructed with CRSP monthly returns from 1966 to 2022. For each stock in month t , the recency score, sign score, cluster score, and return standard deviation are computed using returns both from month $t - 12$ to $t - 1$ (to obtain the updated three feature scores) and from month $t - 18$ to $t - 7$ (to obtain the previous three feature scores). Then, stocks are sorted based on the difference between the updated and previous three feature scores (i.e., shocks) within each volatility decile to form high (top 30 percentile) and low (bottom 30 percentile) portfolios for each feature in month t . Logarithmic value weighted returns of high and low in all deciles are used to compute volatility adjusted period feature-return difference as an average across ten deciles, and finally the period feature returns are summed across all past periods to generate the cumulative returns shown on the figure.

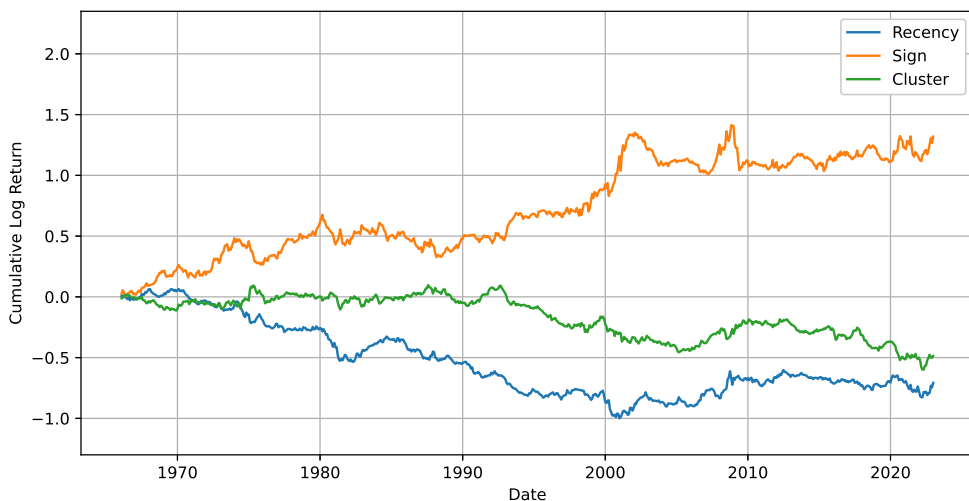
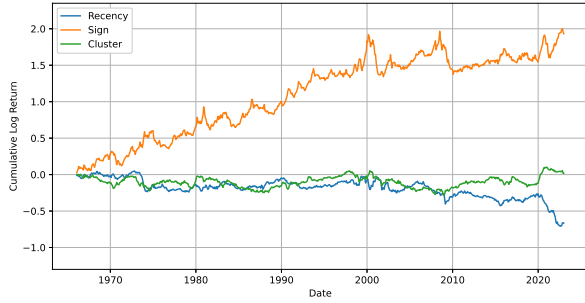
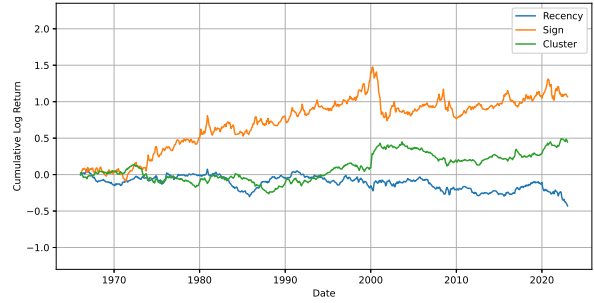


Figure 14: Cumulative Log Returns with Different Lags

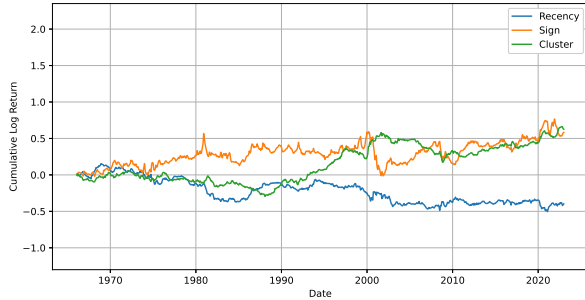
The figure shows the cumulative log returns of high-minus-low portfolios for recency, cluster and sign. The portfolios are constructed with CRSP monthly returns from 1966 to 2022. For each stock in month t , the recency score, sign score, cluster score, and return standard deviation are computed using returns from month $t - 15$ to $t - 4$ for panel a, $t - 18$ to $t - 7$ for panel b, $t - 21$ to $t - 10$ for panel c, and $t - 24$ to $t - 13$ for panel d, respectively. Then, stocks are sorted based on each of the three feature scores within each volatility decile to form high (top 30 percentile) and low (bottom 30 percentile) portfolios for that feature in month t . Logarithmic value weighted returns of high and low in all deciles are used to compute volatility adjusted period feature-return difference as an average across ten deciles, and finally the period feature returns are summed across all past periods to generate the cumulative returns shown on the figure.



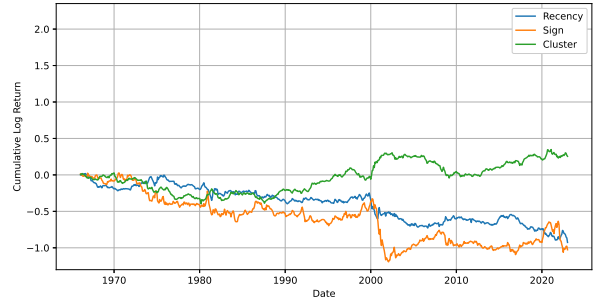
(a) 3-Month Lag



(b) 6-Month Lag



(c) 9-Month Lag



(d) 12-Month Lag